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An Application of Robust Tube MPC to General Anesthesia[★]

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Abstract: Interpatient variability is a main challenge in model-based closed-loop anesthesia, mainly due to poor modeling of peripheral compartments which are weakly informed by clinical data. This paper proposes a tube MPC framework that treats their influence as additive disturbances. Additionally, a new co-administration approach is developed, and a shrinking-horizon strategy is employed to reduce the computational burden of the MPC algorithm. The proposed controller is then evaluated against a baseline PID controller from the literature.

Keywords: Control in system biology, Model predictive control, Applications of optimal control, Biological and pharmaceutical systems, Total intravenous anesthesia, Drug control.

1. INTRODUCTION

During total intravenous anesthesia (TIVA), anesthesiologists adjust drug infusion rates to maintain the depth of hypnosis (DoH) within a suitable range. Closed-loop anesthesia systems have shown promising results in clinical settings (Schiavo et al., 2021). Several strategies have been proposed in the literature, such as PID control for propofol administration (Van Heusden et al., 2013), PID control for propofol and remifentanyl co-administration (Merigo et al., 2019), and model predictive control (MPC) for optimal drug delivery (Aubouin-Pairault et al., 2024).

A common limitation affecting the aforementioned methods is the lack of analytical performance guarantees. Although they have been validated through realistic simulations and clinical trials, formal guarantees can strengthen the trust of clinicians in automated drug delivery systems. In addition, drug co-administration introduces further limitations. The interaction of propofol and remifentanyl creates two degrees of freedom, where multiple drug combinations can yield the same DoH level.

A solution to the first issue was proposed in Fiacchini et al. (2016) where an invariant control law was developed to maintain the evolution of the system indefinitely within a desired subset of the state space. The second issue is addressed by exploiting the structure of the model, such that a clinically relevant target concentration of one drug is selected, and the corresponding equilibrium of the other drug is computed accordingly.

This work presents a novel tube-based MPC scheme that guarantees a desired DoH under constrained operation during anesthesia induction. It also introduces a new co-administration method in which propofol and remifentanyl are dosed according to their equilibrium concentrations.

Although the analysis herein focuses on robustness against interpatient variability in muscle and fat compartments, the proposed framework can accommodate more general disturbance sources. A clinically relevant example is the estimation error introduced when blood and effect-site concentrations are reconstructed from DoH readings. These concentrations are assumed to be measurable in our formulation; however, they can be estimated using available methods in the literature, e.g. Moussa et al. (2023).

The paper is organized as follows. Section 2 recalls the standard anesthesia model and the steps for equilibrium computation and model reduction. Section 3 outlines the control objective and design requirements. Section 4 presents numerical simulations and comparisons. Finally, Section 5 closes the paper with concluding remarks.

2. ANESTHESIA MODEL

Compartmental modeling is used to represent the human body as interconnected compartments, each characterized by specific volumes and clearances. The hypnotic effect is measured using the bispectral index ($BIS \in [0, 100]$). The BIS response to drug administration is modeled using the Response Surface Model (RSM) in Minto et al. (2000), which follows a sigmoidal Hill relationship. The study in Bouillon et al. (2004) identified the RSM parameters through clinical experiments based on the models in Minto et al. (1997); Schnider et al. (1999). In the following study, the Schnider, Minto, and Bouillon models are adopted to describe anesthesia dynamics and drug effect.

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2.1 PK-PD Modeling

Pharmacokinetics (PK) describe drug distribution from the central compartment (blood) to peripheral compartments (muscle and fat) and elimination via the liver and kidneys. Pharmacodynamics (PD) describe the physical effect of drugs on the patient. The first part of the PD model is a linear mapping that captures the delay between the change in drug concentration in blood and its corresponding change in the effect-site (brain). The 4th order PK-PD model for single-drug administration is given as:

$$A_{\text{drug}} = \begin{pmatrix} k_{11} & k_{12} & k_{13} & 0 \\ k_{21} & -k_{21} & 0 & 0 \\ k_{31} & 0 & -k_{31} & 0 \\ k_{e0} & 0 & 0 & -k_{e0} \end{pmatrix}, B_{\text{drug}} = \begin{pmatrix} b \\ 0 \\ 0 \\ 0 \end{pmatrix}. \quad (1)$$

For the case of co-administration, the continuous-time model in (1) is expressed in the following decoupled form:

$$A = \begin{pmatrix} A_p & 0_{4 \times 4} \\ 0_{4 \times 4} & A_r \end{pmatrix}, B = \begin{pmatrix} B_p & 0_{4 \times 1} \\ 0_{4 \times 1} & B_r \end{pmatrix}, \quad (2)$$

which can be compactly written as:

$$\dot{x} = Ax + Bu, \quad (3)$$

where A_p , B_p , A_r , and B_r correspond to the state and input transition matrices of propofol and remifentanyl, respectively. The state vector $x = (x_p, x_r)^\top \in \mathbb{R}^8$, with $x_p \in \mathbb{R}^4$ and $x_r \in \mathbb{R}^4$, represents the concentration of drugs in the blood, muscle, fat, and effect-site compartments, while $u = (u_p, u_r)^\top \in \mathbb{R}^2$ denotes the infusion rates.

Finally, the system in (3) is discretized with a sampling time $t_s = 1$ s. The discrete-time PK-PD model is given by:

$$x^+ = A_d x_k + B_d u_k \quad (4)$$

where A_d and B_d are the discrete-time counterparts of the state and input transition matrices in (2).

2.2 Output Mapping

The second part of the PD model consists of a nonlinear Hill function that maps the effect-site concentrations to the level of brain activity. The BIS signal is modeled as in Bouillon et al. (2004), such that:

$$\text{BIS} = \text{BIS}_0 \left(1 - \frac{U(x_{p4}, x_{r4})^\gamma}{1 + U(x_{p4}, x_{r4})^\gamma} \right), \quad (5)$$

where $\text{BIS}_0 = 97.4$ is the baseline BIS value, $\gamma = 1.43$ is a slope coefficient that describes the steepness of the Hill curve, and $U(\cdot)$ is a drug interaction term defined as:

$$U(x_{p4}, x_{r4}) = \frac{x_{p4}}{C_{50p}} + \frac{x_{r4}}{C_{50r}}. \quad (6)$$

Here, $C_{50p} = 4.47$ and $C_{50r} = 19.3$ are the half-effect concentrations of propofol and remifentanyl, respectively. Note that the Bouillon et al. (2004) model assumes a purely additive interaction between propofol and remifentanyl. This simplification can be relaxed without affecting the validity of the proposed approach. Throughout, let $\text{BIS} \in [\underline{\text{BIS}}, \overline{\text{BIS}}]$ denote the safe surgical range, where $\underline{\text{BIS}}$ and $\overline{\text{BIS}}$ are its lower and upper bounds.

2.3 Equilibrium Computation

Equilibrium is reached when drug concentrations are maintained equal and constant in all compartments. From (3),

the state $x_e = (x_{pe}, x_{re})^\top$ and input $u_e = (u_{pe}, u_{re})^\top$ values at equilibrium can be computed as:

$$\begin{cases} x_{pe} = x_{p1e} = x_{p2e} = x_{p3e} = x_{p4e}, \\ u_{pe} = k_{10p} x_{pe} V_{1p}, \\ x_{re} = x_{r1e} = x_{r2e} = x_{pre} = x_{pre}, \\ u_{re} = k_{10r} x_{re} V_{1r}. \end{cases} \quad (7)$$

To compute x_{pe} and x_{re} , the BIS function in (5) can be inverted to express $U(\cdot)$ as a function of the BIS:

$$U_{bis}(\text{BIS}) = \left(\frac{\text{BIS}_0}{\text{BIS}} - 1 \right)^{1/\gamma}. \quad (8)$$

By comparing the expressions in (6) and (8), we obtain the following equality:

$$\frac{x_{p4}}{C_{50p}} + \frac{x_{r4}}{C_{50r}} = \left(\frac{\text{BIS}_0}{\text{BIS}} - 1 \right)^{1/\gamma}. \quad (9)$$

Since equilibrium computation has two degrees of freedom, the concentrations x_{pe} and x_{re} are interdependent: different pairs can produce the same BIS value. A practical solution is to fix a target for one drug and compute the equilibrium value of the other as follows:

$$x_{p4}(\text{BIS}, x_{r4}) = C_{50p} \left(\left(\frac{\text{BIS}_0}{\text{BIS}} - 1 \right)^{1/\gamma} - \frac{x_{r4}}{C_{50r}} \right), \quad (10a)$$

$$x_{r4}(\text{BIS}, x_{p4}) = C_{50r} \left(\left(\frac{\text{BIS}_0}{\text{BIS}} - 1 \right)^{1/\gamma} - \frac{x_{p4}}{C_{50p}} \right). \quad (10b)$$

For demonstration, a remifentanyl target of $x_{re} = 3$ $\mu\text{g}/\text{ml}$ is selected. From (10a), this corresponds to a propofol target $x_{pe} = 3.61$ mg/ml for a BIS equilibrium $\text{BIS}_e = 50$. These values are used in the simulations in Section 4.

2.4 Model Reduction

The final modeling step is to center the system in (4) around the target equilibrium. Consider only the fast equilibrating compartments (blood and effect-site), and model the effect of the slow compartments (muscle and fat) as an additive disturbance acting on the reduced system:

$$z^+ = A_z z + B_z v + d, \quad (11)$$

where $A_z \in \mathbb{R}^{4 \times 4}$ and $B_z \in \mathbb{R}^{4 \times 2}$ are the submatrices of A_d and B_d corresponding to the fast dynamics. The vectors $z = x_f - x_{f,e}$ and $v = u - u_e$ denote the deviations of the fast-compartment concentrations x_f and the inputs u from their equilibrium values $x_{f,e}$ and u_e , respectively. Similarly, the disturbance term $d = A_w w$ models the effect of the slow compartments, where A_w is the submatrix of A_d that captures this effect. The vector $w = x_s - x_{s,e}$ represents the deviation of the slow-compartment concentrations x_s from their equilibrium values $x_{s,e}$.

Although this separation introduces modeling uncertainty, assuming that the muscle and fat compartments are poorly modeled is clinically reasonable. Tissue concentrations are not measured directly, so they are poorly informed by data and exhibit large interpatient variability.

2.5 Constraint Definition

The state and input variables are constrained within clinically relevant values. The lower bounds correspond

to zero drug concentration prior to the initial infusion, such that $x_{\min} = \mathbf{0}_8$ and $u_{\min} = \mathbf{0}_2$, where $\mathbf{0}_n$ is an n -dimensional zero vector. The upper bounds are chosen as:

$$\begin{cases} x_{\max} = [x_{p1,\max}, x_{pe}, x_{pe}, x_{p4,\max}, \\ \quad x_{r1,\max}, x_{re}, x_{re}, x_{r4,\max}]^\top, \\ u_{\max} = [u_{p,\max}, u_{r,\max}]^\top, \end{cases} \quad (12)$$

where the blood and effect-site compartments are subject to bounds large enough to account for overshoot following the initial bolus, and the muscle and fat compartments are considered too slow for their concentrations to exceed equilibrium values.

For the reduced system in (11), the centered state (z, w) and input (v) bounds are defined as:

$$\begin{cases} z_{\min} = x_{f,\min} - x_{f,e}, & z_{\max} = x_{f,\max} - x_{f,e}, \\ v_{\min} = u_{\min} - u_e, & v_{\max} = u_{\max} - u_e, \\ w_{\min} = x_{s,\min} - x_{s,e}, & w_{\max} = x_{s,\max} - x_{s,e}. \end{cases} \quad (13)$$

The resulting constraint sets are defined as follows:

$$\mathbb{Z} = \{z \in \mathbb{R}^4 \mid z_{\min} \leq z \leq z_{\max}\}, \quad (14a)$$

$$\mathbb{V} = \{v \in \mathbb{R}^2 \mid v_{\min} \leq v \leq v_{\max}\}, \quad (14b)$$

$$\mathbb{W} = \{w \in \mathbb{R}^4 \mid w_{\min} \leq w \leq w_{\max}\}, \quad (14c)$$

where $\mathbb{Z}$, $\mathbb{V}$, and $\mathbb{W}$ are the sets of all admissible z , v , and w values for the system in (11), respectively.

3. CONTROL DESIGN

The objective of this study is to design a control law that is computationally efficient for online closed-loop operation, strictly enforces the physical rate limits of the infusion pumps, safely drives the BIS to its target equilibrium (e.g., $\text{BIS}_e = 50$), and maintains the BIS signal indefinitely within a safe range suitable for surgery, e.g. $\text{BIS} \in [40, 60]$. Model predictive control (MPC) is well-suited to satisfy these requirements as it handles state and input constraints and provides guarantees on asymptotic tracking of BIS_e and containment of BIS within the prescribed region through properly designed terminal cost and constraint set.

For our development in this section, we recall the following set-theoretic concepts. Given two sets $\mathcal{S}, \mathcal{T} \subseteq \mathbb{R}^n$, their Minkowski sum and Pontryagin difference are given by $\mathcal{S} \oplus \mathcal{T} = \{s + t \in \mathbb{R}^n \mid s \in \mathcal{S}, t \in \mathcal{T}\}$, and $\mathcal{S} \ominus \mathcal{T} = \{s \in \mathcal{S} \mid s + t \in \mathcal{T}, \forall t \in \mathcal{T}\}$, respectively. Given a matrix $M \in \mathbb{R}^{m \times n}$, we have $M\mathcal{S} = \{Ms \in \mathbb{R}^m \mid s \in \mathcal{S}\}$.

3.1 Tube-based MPC

The main challenge in meeting the prescribed objectives is the introduction of the disturbance term d in (11), which can drive the system away from its desired trajectory. Although deterministic MPC exhibits a degree of robustness to such disturbances, it cannot formally guarantee robust stability or satisfy safety requirements. To address this limitation, tube-based MPC explicitly incorporates the effects of disturbances into its design by parameterizing the control input as $u_k = \bar{v}_k + K_e e_k$. This allows us to decompose the system into nominal and error components:

$$\bar{z}^+ = A_z \bar{z} + B_z \bar{v}, \quad (15a)$$

$$e^+ = (A_z + B_z K_e)(z - \bar{z}), \quad (15b)$$

where $\bar{z}$ is the disturbance-free nominal trajectory, e is the deviation of the nominal trajectory in (15a) from the real system in (11), and K_e is a feedback gain designed to maintain the real state z in a bounded neighborhood of $\bar{z}$.

Remark 1. While the tube-MPC framework is developed herein to handle additive disturbances arising from poorly controllable slow dynamics, it can be extended to address other clinically and practically prevalent sources of uncertainty.

Upon analyzing (15b), we compute a set that bounds all possible deviations from the nominal trajectory. The original constraints of the nominal system are then tightened using this set. This guarantees that the real trajectory remains contained within a tube around the nominal one.

3.2 Disturbance Set and Constraint Tightening

Since the concentrations of the slowly equilibrating compartments satisfy $w \in \mathbb{W}$, the resulting additive disturbance term d is contained within a compact set:

$$\mathbb{D} = \{d \in \mathbb{R}^4 \mid A_w w_{\min} \leq d \leq A_w w_{\max}\}. \quad (16)$$

For the error dynamics in (15b), the minimal Robust Positively Invariant (mRPI) set, denoted as $\mathcal{E}_\infty$, is the smallest set that satisfies:

$$A_{cl} \mathcal{E}_\infty \oplus \mathbb{D} \subseteq \mathcal{E}_\infty, \quad (17)$$

where $A_{cl} = A_z + B_z K_e$. Intuitively, this set represents all possible accumulated effects of past disturbances and, as such, contains all possible error trajectories. Provided that A_{cl} is Schur-stable, $\mathcal{E}_\infty$ can be expressed as:

$$\mathcal{E}_\infty = \bigoplus_{i=0}^{\infty} A_{cl}^i \mathbb{D}, \quad (18)$$

where each intermediate set $\mathcal{E}_k = \bigoplus_{i=0}^{k-1} A_{cl}^i \mathbb{D}$ is the k -step reachable set representing all possible errors accumulated over k steps. The iteration in (18) only converges to a bounded set under specific conditions (e.g., when A_{cl} is nilpotent). As such, practical computation requires over-approximation methods to find an invariant $\mathcal{E}_\infty$ as in (Blanchini et al., 2008; Kolmanovsky and Gilbert, 1998).

Thereafter, to ensure robust feasibility, the original state and input constraint sets can be tightened as such:

$$\bar{\mathbb{Z}} = \mathbb{Z} \ominus \mathcal{E}_\infty, \quad (19a)$$

$$\bar{\mathbb{V}} = \mathbb{V} \ominus K_e \mathcal{E}_\infty, \quad (19b)$$

and then imposed on the nominal system.

3.3 Terminal Set Computation

To satisfy the pre-defined objectives, a terminal set must be designed to enforce the safe BIS range. This range can be formulated in terms of constraints on the interaction term between the effect-site concentrations. From equations (6) and (8), we obtain the following inequality:

$$U_{bis}(\bar{\text{BIS}}) \leq U(x_{p4}, x_{r4}) \leq U_{bis}(\underline{\text{BIS}}), \quad (20)$$

where any state pair (x_{p4}, x_{r4}) satisfying the above inequality guarantees that the BIS remains within the desired bounds, e.g. $\text{BIS} \in [\underline{\text{BIS}}, \bar{\text{BIS}}]$.

To center the constraint in (20) around the equilibrium, we define a shifted interaction term $V(\text{BIS}) = U_{bis}(\text{BIS}) - U_{bis}(\text{BIS}_e)$. Applying this shift allows one to express the BIS range constraint for the reduced system in (11) as:

$$V(\overline{\text{BIS}}) \leq U(z_{p4}, z_{r4}) \leq V(\underline{\text{BIS}}). \quad (21)$$

The robust terminal set $\overline{\mathbb{Z}}_f$ must then be designed to be invariant under a nominal terminal control law $\bar{v}_N = K_z \bar{z}$, while ensuring feasibility under the tightened constraints $\overline{\mathbb{Z}}, \overline{\mathbb{V}}$, and the bounds in (21), denoted as $\overline{\mathbb{S}}$. This set can be computed by initializing with the intersection $\overline{\mathbb{Z}} \cap \overline{\mathbb{S}}$ and iterating backwards under the terminal control law using standard set propagation techniques.

3.4 Overall MPC Scheme

Based on the above development, we present the following robust MPC scheme:

$$\min_{\mathbf{z}_k, \mathbf{v}_k} \sum_{j=0}^{N-1} (\|\bar{z}_{k+j|k}\|_Q^2 + \|\bar{v}_{k+j|k}\|_R^2) + \|\bar{z}_{k+N|k}\|_P^2 \quad (22a)$$

$$\text{subject to } z(k) \in \bar{z}_{k|k} \oplus \mathcal{E}_\infty \quad (22b)$$

$$\bar{z}_{k+1|k} = A_d \bar{z}_{k|k} + B_d \bar{v}_{k|k} \quad (22c)$$

$$\bar{z}_{k+j|k} \in \overline{\mathbb{Z}} \quad \forall j \in 0, \dots, N-1, \quad (22d)$$

$$\bar{v}_{k+j|k} \in \overline{\mathbb{V}} \quad \forall j \in 0, \dots, N-1, \quad (22e)$$

$$\bar{z}_{k+N|k} \in \overline{\mathbb{Z}}_f, \quad (22f)$$

where $\bar{z}_k := \{\bar{z}_{k+j|k}\}_{j=0}^N$ and $\bar{v}_k := \{\bar{v}_{k+j|k}\}_{j=0}^{N-1}$ are the predicted nominal state and input sequences. Here, $Q \succeq 0$, $R \succ 0$, and $P \succ 0$ is the solution to the discrete-time algebraic Riccati equation. The corresponding terminal control law $\bar{v}_N = K_z \bar{z}$ is the LQR law associated with these matrices. Note that the initial state $\bar{z}_{k|k}$ is not necessarily fixed to the measured state but is treated as a free optimization variable. This is possible because the deviation between the real and nominal states is bounded by the set in (18). As a result, any $\bar{z}_{k|k} \in \overline{\mathbb{Z}}$ that satisfies (22b) ensures that the real state vector remains inside the tube. The resulting feedback law is defined as follows:

$$u(k) = \bar{v}_{k|k}^* + K_e \left(z(k) - \bar{z}_{k|k}^* \right), \quad (23)$$

where $\bar{v}_{k|k}^*$ and $\bar{z}_{k|k}^*$ are the optimal control input and state at the current time step, respectively.

Theorem 1. If the proposed optimization problem in (22) is feasible at step $k = 0$, then it is recursively-feasible and results in control actions ensuring the asymptotic stability of the nominal system.

Proof. The proof follows from the invariance of $\mathcal{E}_\infty, \overline{\mathbb{Z}}_f$, and standard MPC arguments (Mayne et al., 2005). $\square$

Remark 2. The problem in (22) may initially be infeasible due to the size of $\mathcal{E}_\infty$ which prevents the satisfaction of (22b). A common remedy is to temporarily revert to reachability-based tightening of the form $\mathbb{Z} \ominus \mathcal{E}_k$ and to set the initial state as $\bar{z}_{k|k} = z(k)$ until (22b) is satisfied. This improves the initial feasibility of (22) while preserving the properties in Theorem 1 (Chisci et al., 2001).

The proposed formulation assumes a fixed prediction horizon. However, as the state approaches the terminal region,

the influence of uncertainty diminishes and the control objective shifts towards maintaining the state inside the terminal set. To reflect this, a shrinking horizon strategy is adopted (see Algorithm 1), in which the prediction horizon N is gradually reduced until the system enters the terminal region. This simplifies the enforcement of the terminal constraint (22f), reduces the number of decision variables, and thereby reduces the computational burden while preserving the robust feasibility and stability guarantees.

Algorithm 1: Shrinking-Horizon Tube-MPC

```

1:  $N_k \leftarrow N_{\max}$ 
2: for  $k = 0, 1, \dots$  do
3:   if  $\bar{z}_{k|k} \notin \overline{\mathbb{Z}}_f$  then
4:      $N_k \leftarrow \max(N_{\max} - k, N_{\min})$ 
5:   else
6:      $N_k \leftarrow N_{\min}$ 
7:   end if
8:   if  $\bar{z}_{k|k} \in \overline{\mathbb{Z}}$  satisfies (22b) then
9:     Solve tube-MPC (Equation (22))
10:  else
11:     $z_{k|k} \leftarrow z(k)$ 
12:    Solve reachability-MPC (Remark 2)
13:  end if
14:  Apply  $u(k) \leftarrow \bar{v}_{k|k}^* + K_e \left( z(k) - \bar{z}_{k|k}^* \right)$ 
15: end for
```

The condition in line 8 is satisfied when $z(k) \in \overline{\mathbb{Z}} \oplus \mathcal{E}_\infty$.

4. NUMERICAL SIMULATION

For the MPC in Section 3.4, consider the decoupled model in (2) for a 26 year old, 180 cm, 85 kg male patient. The MPC is initialized with maximum and minimum horizon lengths $N_{\max} = 120$ s and $N_{\min} = 30$ s, and the simulation time to $T_{\text{sim}} = 300$ s. The maximum infusion rates are 6.67 mg/s for propofol and 16.67 $\mu\text{g/s}$ for remifentanyl. The upper bounds on the states are set to $x_{\max} = [20 \ 3.6113 \ 3.6113 \ 5.7543 \ 20 \ 3 \ 3 \ 24.8453]^\top$, following the reasoning described in Section 2.5.

Figure 1 shows that the proposed MPC closely mimics the behavior of anesthesiologists: it delivers an initial bolus of both drugs, then gradually decreases infusion rates until convergence to their equilibrium values. Figure 2 shows the state evolution for different projections of the terminal set. Both nominal and real trajectories remain contained within the tube. Once the real trajectory enters $\overline{\mathbb{Z}}_f$, it never leaves, guaranteeing that the BIS evolution remains indefinitely in the desired region.

4.1 Comparison

In this section, the proposed MPC algorithm is compared to a baseline PID controller from the literature (Merigo et al., 2019), using remifentanyl-propofol ratios of $r = 2$ and $r = 0.5$, respectively. PID was chosen as a basis for comparison due to its prevalence in anesthesia research. A population of 50 adults with equal gender distribution was generated, with age 18-64 years, height 162-190 cm, and weight 55-97 kg.

The results in Figures 3 and 4 highlight the advantages of the proposed MPC over the PID approach in meeting

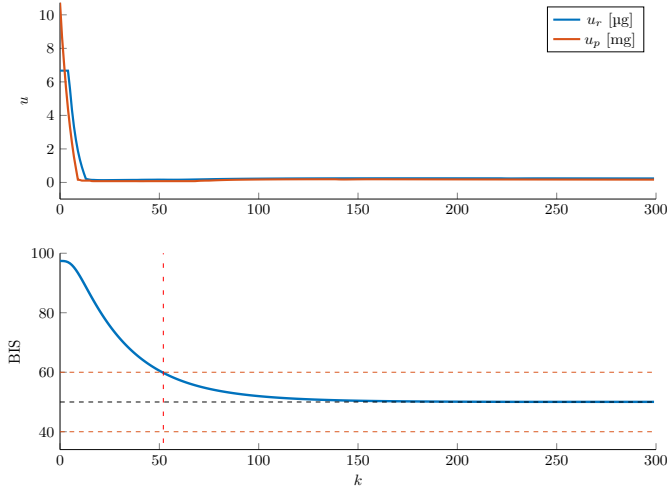


Fig. 1. Nominal patient Input-output evolution. Horizontal orange and black lines denote the safe BIS range and target value. The vertical line marks sleep onset.

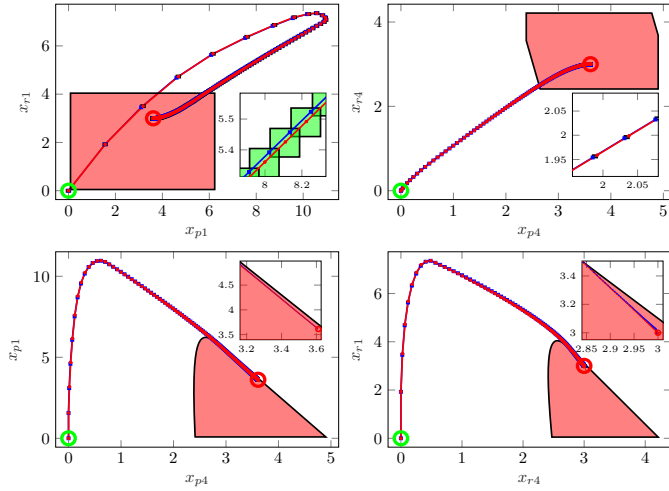


Fig. 2. State evolution with respect to the the terminal set. Red sets show projections of $\mathbb{Z}_f$. Green and red circles mark initial and equilibrium points. Blue and red curves are actual and nominal trajectories. The green region is the tube around the nominal trajectory.

medical requirements. The drug infusion profiles under MPC match anesthesiologists' common practice, while the PID controllers deliver smaller initial dosages and converge more slowly to their corresponding equilibria. The BIS trajectory under MPC also converges to the target value faster and without undershoot, while the PID controllers exhibit a small undershoot and weaker convergence.

In Figure 4, the proposed MPC drives drug concentrations in the blood and effect-site compartments to the desired equilibrium. In contrast, fixed-ratio PID controllers can lead to imbalanced drug dosing: a ratio of 2 leads to excessively high remifentanyl concentrations, while a ratio of 0.5 results in concentrations that can be too low.

The overall performance comparison is summarized in Table 1, reporting the time to sleep (i.e., when $\text{BIS} < 60$), the Integral Absolute Error (IAE($\cdot$)) between the measured and target BIS values, the total drug consumption, and the final blood and effect-site concentrations.

Table 1. Performance comparison summary.

Metric	MPC		PID _{0.5}		PID ₂	
	μ	σ	μ	σ	μ	σ
Sleep time [s]	59.2	1.16	61.9	1.18	73.4	1.14
IAE(BIS - 50)	1964	33.5	2649	8.9	2967	38.4
Total u_p [mg]	111.9	7.51	127.2	7.01	93.9	3.74
Total u_r [µg]	87.8	4.74	63.6	3.50	187.9	7.48
Last x_{p1} [mg/ml]	3.61	0.00	3.92	0.04	2.95	0.09
Last x_{p4} [mg/ml]	3.61	0.00	4.14	0.04	3.13	0.11
Last x_{r1} [µg/ml]	3.00	0.00	2.22	0.22	6.65	0.51
Last x_{r4} [µg/ml]	3.00	0.00	2.24	0.20	6.76	0.45

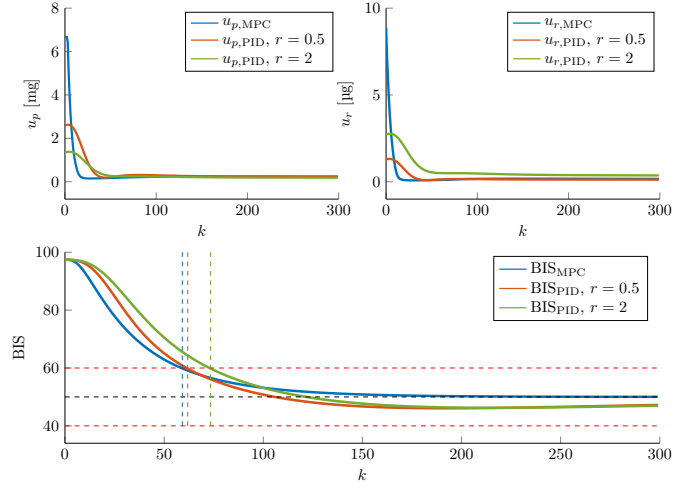


Fig. 3. Average input-output evolution over time. Horizontal red and black lines denote the safe BIS range and target value. The vertical lines mark sleep onset.

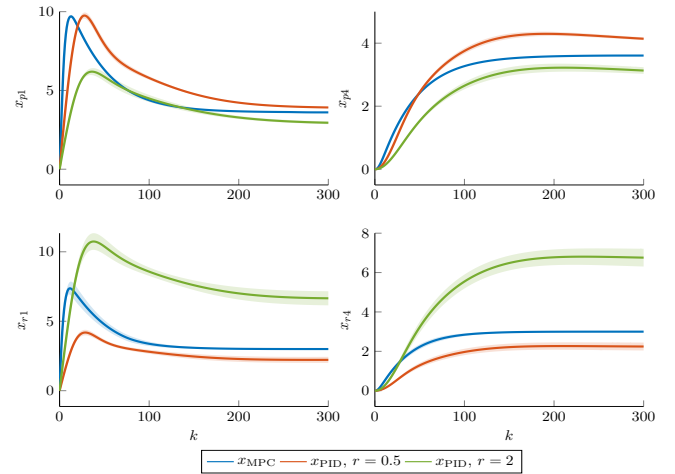


Fig. 4. Time evolution of the mean and standard deviation of states x_{p1} , x_{p4} , x_{r1} , and x_{r4} .

4.2 Discussion

The tube MPC framework provides built-in guarantees of recursive feasibility, constraint satisfaction, and robust stability for all admissible uncertainties, which are valuable for ensuring safety in automated drug delivery. Nevertheless, PID-based methods remain attractive due to their ability to handle model mismatches. However, as more accurate anesthesia models emerge (Eleved et al., 2017,

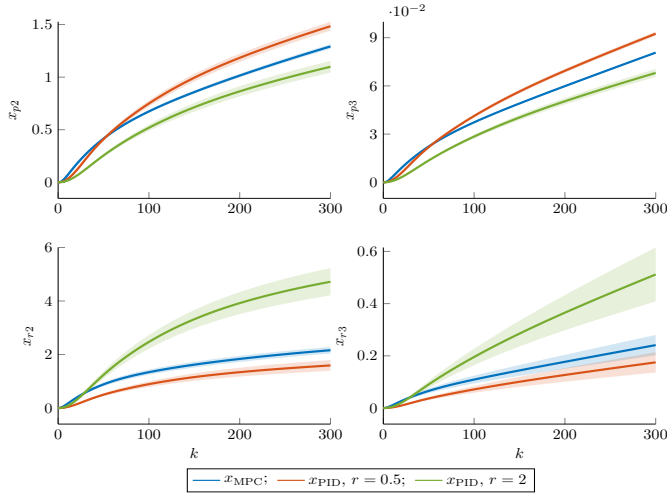


Fig. 5. Time evolution of the mean and standard deviation of states x_{p2} , x_{p3} , x_{r2} , and x_{r3} .

2018), model-based methods are becoming more appealing as performance can be guaranteed analytically. In addition to bounding the time required to reach the terminal set, the shrinking horizon reduces the computation at each step. On average, the time per MPC step decreases from $t_{\text{step}} = 0.0244$ s to $t_{\text{step}} = 0.0057$ s, a 76.64% reduction. Finally, the shaded region in Figure 5 shows that the proposed MPC framework mitigates interpatient variability from poorly modeled muscle and fat compartments.

5. CONCLUSION

In this work, a robust tube MPC framework was developed to control anesthesia dynamics. The proposed framework is suitable for online implementation, and introduces a new method for co-administration by computing target equilibria and regulating the BIS via the interaction term between brain concentrations. However, the current formulation suffers from two limitations: 1) it assumes blood and effect-site concentrations to be measurable, and 2) it relies on a deterministic model. Future work will address these limitations by considering robust observer design and incorporating modeling, measurement, and state estimation uncertainties, and evaluating the sensitivity of the controller to variations in the model parameters.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this work, the authors used ChatGPT to assist in language refinement. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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